

Activity 01: Analysis of Machine Learning and Deep Learning using GitHub

Copilot

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The rise of artificial intelligence (AI) has led to widespread adoption of such tools in nearly every area of society, from the creative arts to programming. However, many operate AI tools without a clear understanding of the models behind them or the processes that lead to the desired results. This paper will utilize a popular AI code assistant, GitHub Copilot, to explore foundational AI concepts such as machine learning (ML) and deep learning. First, various types of ML models will be listed. Next, the relationship between ML and deep learning will be discussed. Finally, an analysis of the effectiveness of GitHub Copilot in exploring these concepts will be given.

The first type of ML model is supervised learning. Supervised learning utilizes labeled data to predict outcomes (GitHub Copilot, n.d.). Some models, like linear or logistic regression, are relatively simple. Others, such as support vector machines (SVMs), can handle much more complicated datasets with the drawback of being computationally expensive (GitHub Copilot, n.d.). The second type of ML model is unsupervised learning, which has the goal of identifying patterns in unstructured data (GitHub Copilot, n.d.). The most common type of unsupervised learning is k-means clustering, which separates data into k classes, where k is a predetermined integer (GitHub Copilot, n.d.). The final type of ML that will be discussed here is reinforcement learning. This approach involves assigning reward or punishment values based on the actions taken and the corresponding reactions from the environment. (GitHub Copilot, n.d.). Over time, the model attempts to learn the best actions for a given situation by seeking the actions that result in the greatest reward.

While ML models provide the foundation for many AI applications, deep learning offers a more advanced approach that allows for handling larger datasets with more complex relationships. Deep learning is closely related to ML, with deep learning representing a

specialized subset of ML (GitHub Copilot, n.d.). Deep learning refers to the use of artificial neural networks (ANNs), modeled off the human brain, with many layers to accomplish a task (Liu et al., 2025). ML typically focuses on smaller, simpler datasets and relies on manual feature extraction. Deep learning, on the other hand, excels at automatically extracting patterns from large and unstructured datasets (GitHub Copilot, n.d.). In this sense, deep learning is typically a form of unsupervised learning (Liu et al., 2025). As a result, deep learning requires significantly more computational power and resources than standard ML models (GitHub Copilot, n.d.).

GitHub Copilot is an AI “Pair Programmer” that utilizes generative pre-trained transformer (GPT) models to assist developers in writing, understanding, debugging, and testing code directly in their development environments (GitHub Copilot, n.d.). Copilot is able to behave as both a chatbot with a dedicated tab and as an agentic AI that directly modifies code. GitHub Copilot was employed to explore the subject of this paper, creating a “meta” scenario where AI is used to understand itself. To begin, Copilot was largely successful in both enumerating various ML models and explaining the relationship between ML and deep learning at a high level. As seen in the accompanying chat logs, Copilot’s responses are easy to digest due to their markdown-style formatting. The content is well-founded and agrees with information from Liu et al., 2025.

However, there are some noticeable drawbacks to using Copilot in this manner. One obvious limitation of Copilot is that its model is tailored to be a code assistant. This resulted in Copilot’s initial refusal to provide an overview of ML models and deep learning, because the prompt was phrased as a research question rather than code generation or alteration. After relating the original questions to software engineering, Copilot had no further qualms about answering the prompts. Another potential downside to using Copilot or any AI model to answer

questions such as these is the relevancy of their data. As noted in the final chat response, Copilot's training data stops after October 2023. Finally, the information provided is accurate but would also be considered quite general knowledge. For example, Copilot does not mention how ANN layers work, with weights, activation functions, and training epochs. With further prompting it is likely that more specific detail could be uncovered. Having explored GitHub Copilot's strengths and limitations, it is time to reflect on its role in understanding AI as a whole.

In conclusion, there are a variety of ML models suited to different generative and classification tasks. Deep learning is a subset of ML, and Copilot serves as an effective tool for gaining a general overview of these topics. In addition to providing insights into ML models, GitHub Copilot demonstrates the potential of AI as an educational tool. By acting as both a guide and a hands-on assistant, Copilot could revolutionize methods for learning complex topics, making AI-driven education more interactive and accessible.

References

GitHub Copilot. (n.d.). <https://github.com/features/copilot/ai-code-editor>

Liu, A. C., Law, O. M., & Law, I. (2025). *Machine Learning, Data Science Foundations with Python*. zyBooks, a Wiley brand. <https://learn.zybooks.com/zybook/CS-530-10427.202617-1>